Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/23/2008
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Project Title:	Enhancement of Pool Boiling Heat Transfer and Control of Bubble Motion in Microgravity using Electric Fields-BCOEL		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	12/01/2001	**End Date:**	11/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NCE to 11/30/2006 per PI (5/2006)		
Key Personnel Changes/Previous PI:	During 2006, Dr. Jungho Kim from the University of Maryland, College Park joined the team. Boiling experiments on the microheater array under the influence of electric fields were conducted in his lab. The participation of Dr. Estelle Iacona was less than planned because of her responsibilities as Assistant Professor/Professor at the Ecole Centrale in Paris. She accepted this position during the first project year.		
COI Name (Institution):	Iacona, Estelle (Johns Hopkins University)		
Grant/Contract No.:	NAG3-2726		
Performance Goal No.:			
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Task Description:	A combined multidisciplinary theoretical and experimental approach was taken to study the bubble formation, detachment and motion as well as heat transfer enhancement in boiling under microgravity. Previous theoretical work was extended to to accommodate nonuniform electric fields, determine the magnitude of the electric field force components, predict bubble detachment volumes under different experimental conditions, predict bubble trajectories in the liquid and explore possible strategies to control bubble formation, size, coalescence and trajectories. The experimental investigation focuses on visualizing bubble formation, bubble shape, and measurement of temperature and heat flux on the heated surface, using high speed cinematography and data acquisition. Heat fluxes were measured and boiling curves weregenerated for prescribed temperature boundary condition at various subcooling levels. The magnitude of the electric field was varied and different electrode configurations were used. Bubble injection was used in the bubble control experiments and force control measurements. Injected bubbles was also used in the study of bubble trajectories and their control by means of electric fields in 1g. - Multiple (few) bubbles were formed by boiling and injection to study the interaction of bubbles, their coalescence with and without electric fields and the influence of coalescence on heat transfer in 1g experiments. The experimental hardware for the experiments allowing the formation of multiple bubbles by injection was designed during the first project year. - The behavior of a single air bubble, its dynamics and interaction with the thermal boundary layer under various electric field configurations was studied. - During the first project year experiments were carried out under terrestrial conditions for different bubble injection regimes (wide range of air flow rates and Weber numbers) to study the influence of the electric field on bubble formation under a variety of conditions. - The analysis of experimental data from low gravity experiments on board of the KC-135 continued. The analysis focused on the influence of the gravity level (microgravity, Lunar gravity, Martian gravity as well as 1g) on bubble detachment under the influence of electric fields. In the nucleate boiling regime, high heat fluxes can be achieved driven by relatively small temperature differences between the solid surface and the bulk fluid. Therefore, pool boiling heat transfer is important in the development of efficient cooling schemes in microgravity applications. The objective of this task is to i) explore the physical mechanisms responsible for the electrohydrodynamic (EHD) heat transfer enhancement in pool boiling in low gravity conditions, ii) quantify the magnitude of enhancement, iii) achieve better control of the boiling process, iv) influence bubble formation, and v) influence bubble motion. Understanding of the mechanisms that affect the heat transfer is crucial for practical applications where pool boiling is needed. The effects of introducing the electric field on heat transfer is also crucial from an industrial standpoint. Example of issues this research will help answer are: whether strong electric fields can prevent coalescence of bubbles, and whether the applied fields can improve and control the bubble diameter on detachment. Better understanding of the effect of electric fields on boiling in microgravity and our ability to predict heat transfer rates under these conditions will open new avenues for aerospace technology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding of the mechanisms that affect the heat transfer is crucial for practical applications where pool boiling is needed. The effects of introducing the electric field on heat transfer is also crucial from an industrial standpoint. Example of issues this research will help answer are: whether strong electric fields can prevent coalescence of bubbles, and whether the applied fields can improve and control the bubble diameter on detachment. Better understanding of the effect of electric fields on boiling in microgravity and our ability to predict heat transfer rates under these conditions will open new avenues for aerospace technology.
	Bubble formation, detachment, coalescence and trajectories under the influence of electric fields The formation and detachment of air bubbles injected into a stagnant, isothermal liquid through an orifice in a plane surface were studied under variable reduced (lunar, martian, microgravity) and terrestrial gravity levels under the influence of both uniform and nonuniform electric fields. The analysis of experimental data from low gravity experiments on board of the KC-135, conducted during the previous project period, continued. The objective of this study is to investigate and explore the possibility of substituting the buoyancy force with the electric field force in aerospace applications by applying an external electric field. Two top electrode geometries were used: flat (uniform electric field) and off-axis spherical (nonuniform electric field). For each of these electrodes, experiments were carried out under variable gravity level and electric potential differences. Bubble life cycle was first experimentally visualized, and then the characteristic dimensions of the bubble and its volume at the moment of detachment were measured using an image processing code. It was shown that both the level of gravity and the magnitude of the electric field significantly affect bubble formation and detachment. Under the influence of the electric field bubbles were elongated and detached regularly in microgravity conditions. It was found that the spherical top electrode deforms the bubble shape and in turn deflects its ascending trajectory. Furthermore, it was observed that the bubble can be tilted in the directions both towards and away form the top electrode under the influence of non-uniform electric fields even during the same one experimental run. Multiple (few) bubbles were formed by injection, to study the interaction of bubbles, their coalescence with and without electric fields and the influence of coalescence on heat transfer in 1g experiments. The experimental hardware for the experiments allowing the formation of multiple bubbles by injection was designed and incorporated into the experimental hardware developed during previous project period. Experiments were carried out under terrestrial conditions for different bubble injection regimes (wide range of air flow rates and Weber numbers) to study the influence of the electric field on bubble formation under a variety of conditions. The results of experimental visualization clearly indicate the significant differences in bubble shape, size and trajectory due to the effects of electric fields. In single bubble experiments the bubble can be tilted towards or away from the upper electrode under the influence of electric field. Although the direction of the tilt can alternate (even in a single experiment), in the majority of the cases it is towards the electrode. In dual bubble experiments the bubbles behave similar to the single bubble case at larger volume flow rates. In low volume-flow-rate experiments the bubbles can tilt in different directions. Furthermore, the detachment volume of the bubble decreases under the influence of electric fields, which leads to a higher detachment frequency. The use of the electric field has been proven to be a promising way to control bubble behavior and motion and assist the process of bubble detachment in microgravity. Key conclusions

Task Progress:

• Bubble volume decreases and the elongation increases with increasing electric field magnitude

• The detachment contact angle changes

• Bubbles move away from the electrode at which they have formed

• Bubble coalescence is less pronounced

• Regular bubble detachment is observed in microgravity under the influence of electric fields, contrasted to irregular bubble shape, behavior and detachment without electric fields.

• The influence of electric fields on gas-liquid two-phase systems is more pronounced in microgravity than in terrestrial conditions.

• The bubble volume at detachment and detachment time decreased considerably as the gravity level increased (from microgravity, through lunar and martian gravity conditions). No appreciable change on bubble shape and formation was observed.

• An initially uniform electric field elongates the bubble in the direction of the electric field lines. When increasing the applied electric field magnitude bubble volume at detachment decreases, bubble aspect ratio at detachment increases.

• The effect of the electric field is more pronounced under microgravity conditions than in terrestrial conditions or reduced gravity conditions (lunar, martian).

• For the investigated electric field magnitude, the impact of electric potential difference on bubble detachment is not as pronounced as the variation of gravity.

• In terrestrial conditions a nonuniform electric field considerably modifies the bubble shape and trajectory during formation, detachment and ascent into the liquid.

• In single bubble experiments in a nonuniform electric field the bubble was tilted towards or away from the upper electrode under the influence of the electric field. Although the direction of the tilt alternated (even in a single experiment), in the majority of the cases it tilted towards the electrode.

• In dual bubble experiments the bubbles behaved similarly to the single bubble case at larger volume flow rates. In low and moderate volume flow rate experiments the bubbles are tilted in opposite directions under the influence of electric fields.

Recommendations for future work

Work currently underway and recommended for future studies involves the following aspects of bubble formation and motion: • Computational modeling and determination of the magnitude of electric force components acting on the bubble.

• Coalescence with multiple bubbles (3 and 4) to better understand the effects of coalescence.

• Coalescence using different working fluids to better understand the properties of the working fluid.

• Coalescence experiments in microgravity to investigate if the presence of the electric field is going to affect the pronounced coalescence behavior in microgravity.

• Study of different electrode configurations for single and multiple bubble injection and boiling experiments

• A high-quality g-level as well as longer duration of the microgravity environment is desirable in the experiments with bubbles.

Boiling under the influence of electric fields

In the nucleate boiling regime, high heat fluxes can be achieved driven by relatively small temperature differences between the solid surface and the bulk fluid. Understanding of the mechanisms that affect the heat transfer is crucial for practical applications where pool boiling is needed. The effects of introducing the electric field on heat transfer is also crucial from an industrial standpoint. Example of issues this research will help answer are: whether strong electric fields can prevent coalescence of bubbles, and whether the applied fields can improve and control the bubble diameter on detachment. Pool boiling heat transfer is crucial in the development of efficient cooling schemes in microgravity applications. Better understanding of the effect of electric fields on boiling in microgravity and our ability to predict heat transfer rates under these conditions will open new avenues for aerospace technology.

The objective of this task was to explore the physical mechanisms responsible for the electrohydrodynamic (EHD) heat transfer enhancement in pool boiling in low gravity conditions, quantify the magnitude of enhancement, achieve better control of the boiling process, influence bubble formation and motion. The experimental investigation focused on visualizing bubble formation, bubble shape, and measurement of temperature and heat flux on the heated surface, using high speed cinematography and data acquisition. Heat fluxes were measured and boiling curves were generated for prescribed temperature boundary condition at various subcooling levels. The magnitude of the electric field was varied and different electrode configurations were used. Results of these investigations provide valuable insights for the design of phase change equipment for microgravity applications. Better understanding of the effect of electric fields on boiling in microgravity and our ability to predict heat transfer rates under these conditions will open new avenues for aerospace technology.

The effects of a nonuniform electric field on vapor bubble nucleation, development and detachment and the associated change of the heat transfer rate were studied in subcooled pool boiling at earth gravity (upward facing heater) and a simulated -1 gravity (downward facing heater) using a microscale heater array maintained at constant temperatures. In the experiments the boiling rig developed by Dr. Jungho Kim at the University of Maryland, College Park was used. The effects of a nonuniform electric field on vapor bubble detachment and heat transfer rate were studied in pool boiling at different subcooled conditions for various wall temperatures. Dielectric fluid (FC-72) was used as the working fluid at 1 atm at earth gravity with two extreme gas concentration levels. An array of independently controlled microheaters each mm2 in size were maintained at constant temperature using electronic feedback loops, enabling the heat transfer from each heater to be determined. An electric field was applied between the horizontal upward facing microheater

array, which was grounded, and a spherical, off-axis top electrode. Boiling heat transfer data with and without the electric field were obtained in this study. Without the electric field, a single large ‘‘primary’’ bubble was observed to form due to the coalescence of the individual “satellite” bubbles which nucleated directly from each single heater array. Before its detachment, a dry spot formed underneath this primary bubble resulted in a reduction in heat transfer. With the electric field applied, three or more small “secondary” bubbles, that nucleated and grew more rapidly and detached more frequently, were observed. Due to the nonuniformity of the electric field, bubbles moved away from the top electrode (into the weaker region of the electric field) during their development. Higher overall heat transfer rates were measured from the heater array. In addition, the bubble behavior showed agreement with our previous investigation of injecting air bubbles into a stagnant, isothermal liquid through orifices.

Under the applied nonuniform electric field, the results showed moderate heat transfer enhancement under earth gravity conditions, as indicated by more rapid bubble nucleation and growth and more frequent detachment from the heater surface. When the heater array was facing downward, much larger heat transfer rates were measured. No bubble detachment was observed without the aid of the electric field whereas bubbles detached regularly when the electric field was applied. Due to the nonuniformity of the electric field, bubbles migrated away from the top electrode during their development, which is in agreement with the theory.

Suggestions for future work

• Perform the 3×3 array experiments with and without the electric field in microgravity. Compare the microgravity results with the -1g results.

• Perform experiments, with and without the electric field, with different heater array sizes in earth gravity, microgravity, and -1g gravity conditions.

• Use different shapes of the top electrode and vary the magnitude of the electric field.

• Model the nonuniform surface charge on the bubble-liquid interface and the bubble dynamic behavior.

The results and conclusions of this research are directly relevant for and applicable to terrestrial systems, in particular because most of the experiments were conducted in terrestrial gravity conditions. Bubble control and heat transfer can be achieved successfully in terrestrial conditions. The results for the downward facing surface orientation are particularly significant and dramatic in terms of heat transfer enhancement for terrestrial systems.

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